

Cloud Computing Assignments — Concepts Explained from Basics

1. What is Cloud Computing?

Cloud Computing means using computing resources such as:

- Servers
- Storage
- Databases
- Networking
- Software
- Applications
- Processing power

through the internet instead of managing everything on your own computer.

Instead of buying powerful hardware and maintaining it yourself, cloud providers give these resources on demand.

Examples:

- Google Drive → cloud storage
 - Gmail → cloud-based email service
 - Netflix → cloud-hosted streaming
 - AWS (Amazon Web Services) → cloud platform
 - Microsoft Azure → cloud platform
 - Google Cloud → cloud platform
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2. Why Cloud Computing is Needed

Before cloud computing:

- Companies had to buy expensive servers
- Maintain hardware manually
- Handle power, cooling, networking
- Upgrade systems frequently

Problems:

- High cost

- Difficult maintenance
- Limited scalability
- Hardware failures

Cloud computing solved these problems.

Advantages:

- Pay only for what you use
 - Access resources anytime
 - Easy scaling
 - High availability
 - Faster deployment
 - Remote accessibility
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3. Types of Cloud

A. Public Cloud

Services are available publicly over internet.

Example:

- AWS
- Google Cloud
- Microsoft Azure

B. Private Cloud

Cloud infrastructure used only by one organization.

More secure but expensive.

C. Hybrid Cloud

Combination of public and private cloud.

4. Cloud Service Models

These are very important because all assignments are based on them.

A. IaaS (Infrastructure as a Service)

Cloud provider gives:

- Virtual machines
- Storage
- Network

User manages:

- OS
- Applications
- Data

Example:

- Amazon EC2

Used in Assignment 1.

B. PaaS (Platform as a Service)

Provider gives:

- Platform
- Runtime environment
- Development tools

User only develops applications.

Example:

- Google App Engine
- Salesforce Platform

Used in Assignment 2 and 3.

C. SaaS (Software as a Service)

Complete software provided through internet.

Example:

- Gmail
- Google Docs

- Salesforce CRM

Used in Mini Project assignment.

5. Virtualization

Cloud computing mainly works because of virtualization.

What is Virtualization?

It means creating virtual versions of:

- Servers
- Operating systems
- Storage
- Networks

One physical computer can run multiple virtual machines.

Example: A single powerful server can host:

- Linux VM
- Windows VM
- Database VM

Benefits:

- Better resource utilization
 - Isolation
 - Easy management
 - Scalability
-

6. Important Cloud Terms

Instance

A virtual server in cloud.

VM (Virtual Machine)

Software-based computer.

Scalability

Ability to increase/decrease resources.

Elasticity

Automatic scaling based on demand.

Data Center

Physical location containing cloud servers.

Region

Geographical area where cloud services are hosted.

Availability Zone

Separate data center inside a region.

Assignment 1

Case Study on Amazon EC2 and Learn About Amazon EC2 Web Services

1. What is AWS?

AWS stands for Amazon Web Services.

It is a cloud computing platform provided by Amazon.

AWS provides:

- Servers
 - Databases
 - Storage
 - Networking
 - AI services
 - Security services
-

2. What is Amazon EC2?

EC2 stands for Elastic Compute Cloud.

It provides virtual servers called EC2 instances.

It belongs to:

IaaS (Infrastructure as a Service)

because AWS provides infrastructure.

3. Why EC2 is Used

Instead of buying physical servers:

- Create virtual server instantly
 - Choose CPU/RAM/storage
 - Start and stop anytime
 - Pay only for usage
-

4. Concepts Used in EC2

A. EC2 Instance

Virtual server.

Example: Ubuntu Linux machine running on cloud.

B. AMI (Amazon Machine Image)

Template used to launch instance.

Contains:

- Operating system
- Software
- Configuration

Example:

- Ubuntu AMI
 - Windows AMI
-

C. Key Pair

Used for secure login.

Contains:

- Public key
- Private key

Private key (.pem file) is used for SSH connection.

D. Security Group

Acts like firewall.

Controls:

- Incoming traffic
- Outgoing traffic

Example: Allow:

- SSH (port 22)
 - HTTP (port 80)
-

E. Elastic IP

Static public IP address.

F. EBS (Elastic Block Store)

Storage attached to EC2 instance.

5. Working of EC2

Step-by-step:

1. Login to AWS
 2. Launch EC2 instance
 3. Select AMI
 4. Select instance type
 5. Configure storage
 6. Configure security group
 7. Launch instance
 8. Connect using SSH
-

6. Instance Types

Different configurations available:

General Purpose

Balanced CPU and RAM.

Compute Optimized

High CPU performance.

Memory Optimized

Large RAM.

Storage Optimized

Large storage performance.

7. Benefits of EC2

- Fast deployment
- Scalability
- High availability
- Cost effective
- Flexible

8. Real-Life Example

Suppose a website suddenly gets huge traffic.

Using EC2:

- Add more instances quickly
- Handle more users
- Remove extra servers later

This is called elasticity.

Assignment 2

Installation and Configuration of Google App Engine

1. What is Google Cloud Platform?

Google Cloud Platform (GCP) is Google's cloud service platform.

Provides:

- Compute services
 - Databases
 - Storage
 - AI services
 - Application hosting
-

2. What is Google App Engine?

Google App Engine (GAE) is a:

PaaS (Platform as a Service)

service.

It allows developers to deploy applications without managing servers.

3. Main Idea of App Engine

Developer focuses only on:

- Writing code
- Deploying application

Google handles:

- Servers
 - Scaling
 - Load balancing
 - Maintenance
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4. Concepts Used

A. Deployment

Uploading application to cloud.

B. Runtime Environment

Environment where application runs.

Examples:

- Python
 - Java
 - Node.js
-

C. Scaling

Automatically increasing resources.

D. Load Balancing

Distributing traffic across servers.

5. Standard Environment vs Flexible Environment

Standard Environment

- Faster
- Limited customization
- Cost effective

Flexible Environment

- More customization
 - Uses Docker containers
 - More control
-

6. Steps in Assignment

Step 1

Create Google Cloud account.

Step 2

Install Google Cloud SDK.

Step 3

Initialize SDK.

Command:

```
gcloud init
```

Step 4

Create App Engine project.

Step 5

Write application.

Example: Simple Python Flask app.

Step 6

Create app.yaml configuration file.

Step 7

Deploy application.

Command:

```
gcloud app deploy
```

Step 8

Open application.

Command:

```
gcloud app browse
```

7. Benefits of App Engine

- No server management
- Automatic scaling
- Easy deployment
- Highly available
- Good for web applications

8. Difference Between EC2 and App Engine

Feature	EC2	Google App Engine
Type	IaaS	PaaS
Server Management	User manages	Google manages
Flexibility	High	Moderate

Feature	EC2	Google App Engine
Complexity	Higher	Easier
Main Use	Virtual servers	Application hosting

Assignment 3

Creating an Application in Salesforce.com Using Apex Programming Language

1. What is Salesforce?

Salesforce is a cloud-based CRM platform.

CRM means:

Customer Relationship Management.

Used for:

- Customer management
 - Sales tracking
 - Marketing
 - Business applications
-

2. Salesforce is Based on Which Cloud Model?

Primarily:

- SaaS

Also provides:

- PaaS through Salesforce Platform
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3. What is Apex?

Apex is Salesforce's programming language.

It is similar to Java.

Used for:

- Business logic
 - Database operations
 - Automation
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4. Important Salesforce Concepts

A. Object

Similar to database table.

Example:

- Customer table
 - Employee table
-

B. Record

Single row in table.

C. Field

Column in table.

D. SOQL

Salesforce Object Query Language.

Used to retrieve data.

Similar to SQL.

Example:

```
SELECT Name FROM Student
```

E. Trigger

Code executed automatically on events.

Example:

- Before insert
 - After update
-

F. Visualforce

Framework for UI pages.

5. Steps in Assignment

Step 1

Create Salesforce developer account.

Step 2

Login to Salesforce org.

Step 3

Create custom object.

Example: Student object.

Step 4

Add fields.

Example:

- Name
- Roll Number
- Marks

Step 5

Write Apex class.

Example: Insert records.

Step 6

Run code using Developer Console.

Step 7

Display records.

6. Example Apex Code

```
Student__c s = new Student__c();  
s.Name = 'Aayushi';  
insert s;
```

Explanation:

- Create object
 - Assign value
 - Insert into database
-

7. Why Salesforce is Popular

- Fully cloud based
 - No infrastructure management
 - Business automation
 - Scalable
 - Secure
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Assignment 4

Design and Develop Custom Application (Mini Project) Using Salesforce Cloud

1. Objective

Develop a real-world cloud application using Salesforce.

This assignment combines:

- Database concepts
 - Cloud computing
 - Salesforce development
 - Automation
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2. Concepts Used

A. Cloud Application

Application hosted on cloud.

Accessible through internet.

B. Custom Object

User-created database table.

C. Workflow Automation

Automatic execution of tasks.

D. Reports and Dashboards

Data visualization tools.

3. Example Projects

Student Management System

Features:

- Add student
- Track attendance
- Store marks

Hospital Management

- Patient records
- Doctor schedules
- Appointment tracking

Inventory Management

- Product tracking
 - Stock management
 - Sales reports
-

4. Typical Project Structure

Frontend

Forms and pages.

Backend

Apex classes and logic.

Database

Salesforce objects.

5. Development Steps

1. Define project problem
2. Create objects

3. Create relationships
 4. Add fields
 5. Write Apex logic
 6. Create forms/pages
 7. Test application
 8. Generate reports
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6. Important Salesforce Relationships

Lookup Relationship

Loose connection between objects.

Master-Detail Relationship

Strong parent-child relationship.

7. Skills Learned

- Cloud application development
 - Database design
 - Automation
 - SaaS platform usage
 - Business application design
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Assignment 5

Setup Your Own Cloud for SaaS Over Existing LAN

1. Objective

Create a private cloud-based software service inside laboratory network.

This assignment focuses on:

- SaaS
- Networking
- Web hosting

- Local cloud setup
-

2. What is SaaS?

Software delivered through network/browser.

Users do not install software locally.

Example:

- Google Docs
 - Gmail
 - Office 365
-

3. What is LAN?

LAN means:

Local Area Network.

Computers connected within limited area.

Example:

- College lab
 - Office network
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4. Concepts Used

A. Server

Central computer providing services.

B. Client

Computer using services.

C. Web Hosting

Hosting application on server.

D. Networking

Communication between systems.

5. How SaaS Works in LAN

1. One computer acts as server
 2. Application hosted on server
 3. Other systems access through browser
 4. All users use same application
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6. Technologies Commonly Used

- Apache server
 - XAMPP
 - PHP
 - MySQL
 - Node.js
 - Django
-

7. Example Mini Project

Online Notes Sharing System

Server stores:

- Notes
- PDFs
- User data

Clients access through:

- Browser
 - Local IP address
-

8. General Steps

Step 1

Setup server system.

Step 2

Install web server.

Step 3

Create application.

Step 4

Connect database.

Step 5

Find local IP address.

Example:

```
ipconfig
```

Step 6

Access application from other systems.

Example:

```
http://192.168.1.5
```

9. Skills Learned

- SaaS architecture
- Networking basics
- Server setup

- Web deployment
 - Multi-user systems
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Overall Connection Between All Assignments

Assignment	Cloud Model	Main Technology
Amazon EC2	IaaS	AWS
Google App Engine	PaaS	Google Cloud
Salesforce Apex	PaaS/SaaS	Salesforce
Salesforce Mini Project	SaaS/PaaS	Salesforce
SaaS over LAN	SaaS	Local Cloud

Final Understanding Flow

1. Learn cloud basics
2. Learn infrastructure cloud (EC2)
3. Learn platform cloud (Google App Engine)
4. Learn cloud application development (Salesforce)
5. Learn complete SaaS deployment

This sequence gradually teaches:

- Infrastructure
- Platforms
- Application development
- Deployment
- Real cloud systems